RESEARCH PAPER

Data Analytics & Business Intelligence

Sales Trend and Time-Based Performance Analysis
for Afficionado Coffee Roasters

A Comprehensive Analysis of Transaction-Level Data,
Temporal Demand Patterns & Multi-Location Performance

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Abstract

The report herein provides a thorough investigation into the rate of sales and performance of Afficionado Coffee Roasters, a coffee retailer with many retail locations. Transaction-level data will be analysed using advanced data analytic processes such as temporal feature engineering, statistical aggregates and interactive data visualisations to reveal previously undiscovered

patterns in consumer purchase behaviour by time of day, days of the week and across the various retail locations.

Through this analysis of the consumer purchasing behaviours, the following demand cycles were identified: there are significantly more purchases during morning hours than during any other time of the day for all retail locations. There is also a stark contrast between how consumers behave on weekends compared to weekdays. The retail location performance will have specific signatures. The findings are presented visually via an interactive Streamlit dashboard which enables real-time filtering, drill-down exploration and managerial support for decision making.

Some of the most important recommendations offered here are that specialty coffee retailers should consider implementing time differentiated staffing models, demand aligned scheduling of inventories, promotional calendars that match the customer locations from which they originate, and continuous monitoring infrastructure to their operations process. The evidence provided in this article reveals that employing systematic time based analytic methodologies can lead to tangible improvements across operational efficiencies, increased customer satisfaction and revenue optimization opportunities for specialty coffee retailers.

Key terms: Coffee retailer sales trend analysis (analytics), time based analytics, peak demand forecasting, and transactional data analysis using a Streamlit dashboard, and retail business intelligence; temporal feature engineering, cross location performance analysis, customer behavioral patterns.

1. Introduction

1.1 Background and Context

Over the last decade, the specialized coffee segment of the international retail coffee industry has changed dramatically, as customers have become more and more used to experiencing specialty coffee as a uniquely personal and exceptionally high-quality experience. Afficionado Coffee Roasters straddles the two worlds of artisanal creation and retail scale, serving a wide range of customer segments across multiple metropolitan areas. In this competitive market, data-based decision making is not just a competitive advantage; it has become an absolute necessity in order to operate efficiently.

Food and beverage retail businesses are facing compounding challenges, including fluctuating demand that varies depending on the time of day, the day of the week, the season, and local events; inventory with a short shelf life; and payroll costs that are directly related to the ability to efficiently create employee schedules. Although businesses in this sector capture vast amounts of transactional data through point of sale systems, very often this information goes under-utilized, as it sits in databases with little or no translation into systematic operational intelligence for the business.

1.2 Research Motivation

- Afficionado Coffee Roasters has detailed data about every customer purchase, including the time, location, amount, and item of purchase. This data can be used to create accurate demand curves and patterns of customer behavior. There are three key reasons for this research:
- Existing reports use an aggregated level of data so that daily or weekly fluctuations in demand are lost; these types of demand fluctuations are important for staffing and inventory purposes.
- Store managers do not have a real-time and unified interface to evaluate their store's performance versus the performance of the entire chain.
- Currently, there is a single, uniform promotional/operational strategy across all locations even though there is evidence that different locations have significantly different demand patterns.

1.3 Scope of the Study

The scope of this research is the complete transactional dataset of Afficionado Coffee Roasters, including all recorded purchases made in every store during the time frame of this investigation. It includes identification of temporal patterns using three different time granularities (hourly, daily, weekly), behaviour segmentation (weekend vs weekday behaviour) and cross-location performance benchmarking. Finally, this research provides an interactive dashboard that can be used as a platform to facilitate future analytical decision-making.

1.4 Significance

This research is of importance, as it is relevant to three areas of business: managing your people; planning your inventories; developing your marketing. This study offers an academic contribution through a replicable methodology (based on time) that can be used for all specialty food and beverage operators. By combining feature engineering, visualization science, and the business recommendations that come out of these through one analytical framework, this project will be the first of its kind to provide a holistic approach to applied retail intelligence.

2. Problem Statement

2.1 Core Business Challenges

Although numerous transactional data exists, Afficionado Coffee Roasters is presently trying to improve their operational capabilities. There are four interrelated waypoints through which this study plans to provide assistance to this distinguished coffee company

Challenge Area	Operational Impact
Absence of peak-period identification	Staffing for service delays at peak times and underutilized at off-peak times contribute to poor overall performance of stores.
Limited day-of-week behavioral visibility	Consistent scheduling during the week does not take into consideration demand patterns across Mon.-Sun.
No cross-location performance comparison	Identifying top and bottom performing branches is difficult; therefore, sharing information about business practices is difficult.
Time-based demand fluctuation not quantified	Static, assumption-based purchasing plans create over-ordering or stock shortages.

2.2 Problem Framing

The analytics revolves around the question of how to apply temporal analytics to a given multi-unit retail transaction history in order to methodically identify demand by day of week, hour of day, and retail unit, and to build an interactive data visualization platform to make such information continuously available to business decision makers (analysts).

2.3 Research Questions

1. What are the peak and off-peak demand periods at the Afficionado Coffee Roasters?
2. How does customer purchasing behaviour differ according to weekday/weekend - and by unit?
3. Which Afficionado Coffee Roasters have lowest/ highest revenue and volume per transaction, and what periods of time do these units generate the most in transactions?
4. What time-bucket classification captures the demand structure best for operational planning purposes, and will allow Afficionado Coffee Roasters to better manage their operational resources?
5. How do analytical results data visualisation through the analytic dashboard help non-technical members of the Afficionado Coffee Roasters business to extract applicable insights from analytical data?

3. Objectives of the Study

3.1 Primary Objectives

- Conducting a comprehensive investigation into total sales revenue trends throughout the entire observation period.
- Identifying and quantifying peak demand times and how time-bucket performance varies as determined by hourly transaction data.
- Analyzing performance through a day-of-week lens by comparing weekday versus weekend behavior.
- Specifically comparing revenue and transaction volume performance across all locations of Aficionado Coffee Roasters.
- Creating an interactive, stakeholder-ready dashboard to allow for real-time exploration of analytical insights.

3.2 Secondary Objectives

- Creating meaningful temporal features from raw timestamp data that add depth to analysis.
- Determining if revenue and transaction quantity trends align with or diverge from each other in periods of time.
- Identifying low demand times that may present opportunities for promotional or operational optimization.
- Developing a replicable analytical framework for future data and business periods.
- Providing operation-specific and data-driven recommendations that may be addressed by both store level personnel and chain level planners.

4. Dataset Description

4.1 Data Source

The dataset being used in this analysis is comprised of transaction-level records from Aficionado Coffee Roasters' point-of-sale (POS) system. Each record represents one completed transaction and each transaction record was exported from the POS database in a structured tabular format for analytical purposes

4.2 Dataset Schema

Field Name	Data Type	Description
transaction_id	String / Integer	Unique identifier for each purchase transaction
transaction_date	Date	Calendar date on which the transaction occurred
transaction_time	Timestamp (HH:MM)	Exact time of transaction, enabling hourly analysis
store_id	Categorical	Identifier for the store location (branch code)
store_location	String	Human-readable branch name or address
product_id	String	Unique product identifier

product_category	Categorical	Product group (e.g., Espresso, Drip Coffee, Pastry)
product_detail	String	Specific product name and variant description
unit_price	Numeric (Float)	Price per single unit of the product in local currency
transaction_qty	Integer	Number of units purchased in the transaction

4.3 Derived Fields

- The following derived fields were computed using feature engineering and are not in the raw data; however, they are essential for all analysis done in this paper:
- Revenue — gross revenue per transaction calculated by multiplying transaction quantity by the unit price.
- hour of day — represents the granular hourly aggregation by integer value (0 to 23) derived from the timestamp of each transaction.
- day of week — Categorical variable representing the 7 days of the week (Monday through Sunday), created from the date of each transaction.
- Time bucket — ordinally classifies each transaction as being in 1 of the following 4 time buckets: Morning (6 through 11), Afternoon (12 through 4), Evening (5 through 9), or Late (10 through 5).
- is weekend — a binary flag indicating if a transaction occurred on a weekday (Monday through Friday) or weekend (Saturday through Sunday).
- Week number — for weekly aggregation analysis, derived from the date of each transaction is the week number according to the ISO standard.

4.4 Data Volume and Coverage

The data set consists of many months of uninterrupted retail data of retail operations, consisting of thousands of individual transaction records from all Afficionado Coffee Roasters locations. The volume and temporal scope of the data set allows for stable seasonal patterns, day of the week cycles, and demand signatures at the individual location with statistical validity to be determined.

5. Methodology

To ensure the highest level of data quality, rigorous analysis and accurate interpretability of the data, an analytical process is broken down into five separate phases with specified processes outlined below in Sections

5.1 Data Cleaning and Preprocessing

The POS data files were cleaned through an established process prior to any further DLLS analysis being performed on the file. The following procedures were performed:

- Duplicate records removed — Duplicate transactions (Identical transaction records (same transaction ID, time of transaction, and product), were found and removed leaving only one canonical transaction.
- Missing value handling — Transactions with null values in the key fields of price and quantity were reviewed, and transactions where the pricing or quantity was missing were removed; those transactions where the store information could be inferred from context were populated with that inferred store information from the corresponding key fields.
- Validity filter — Transactions with zero or negative values, or data out of the accepted range for either price or quantity were flagged to be reviewed, and were removed if they could not be validated against supporting documentation.
- Timestamp standardization — Transaction_time values were parsed and stored in 24-hour format, and transaction date fields were loaded to ISO 8601 date format for proper sorting and grouping by time.
- Outlier analysis — The statistical outlier analysis was performed on the unit price and transaction quantity data to identify any outlier values. The outlier records were compared to the product master files prior to being loaded or rejected.

5.2 Feature Engineering

Temporal feature engineering is the analytical core of this study. The following features were constructed systematically from the cleaned dataset:

Feature	Derivation Logic	Analytical Purpose
Revenue	Unit_price x transaction_qty	Primary performance metric for all revenue analyses
hour_of_day	datetime.hour from transaction_time	Enables hourly aggregation and demand curve visualization

day_of_week	datetime.weekday() mapped to day name	Supports day-level comparisons and weekday/weekend split
time_bucket	Conditional binning of hour_of_day	Provides operationally meaningful demand period classification
is_weekend	Binary: 1 if Sat/Sun, 0 otherwise	Enables behavioral comparison across schedule segments
week_number	ISO week from transaction_date	Supports weekly revenue aggregation and trend analysis
rev_per_tx	Revenue / transaction count per period	Measures average basket size changes over time

5.3 Analytical Techniques

This research used four analytical procedures in combination across the data set:

- Temporal aggregation: Transactions aggregated over time (hour, day, week or 'time bucket') to create total revenue, transaction count and average basket size for each aggregation level.
- Comparative analysis: Statistical comparison of metrics by day of week, location and/or time period to determine relative performance differences.
- Heatmap visualization: Two-dimensional intensity mapping of transaction volumes and revenues at the hour and day for each location to identify rising or falling trends in compound temporal patterns.
- Trend decomposition: Decomposing the weekly revenue time series into its trend and variance components to identify whether structural growth or decline exists, or if cyclical fluctuation exists year-on-year.

5.4 Tools and Technologies

- Python 3.x - primary programming language used for all data processing and analysis routines.
- Pandas - Data manipulation library used for temporal aggregation, feature extraction and support for tabular analysis.
- NumPy - numerical computing package used for statistical analysis and perform array operations.
- Matplotlib and Seaborn - Static visualization libraries used to create exploratory and publication-quality graphs/charts.

- Plotly - Interactive visualisation package that enables zooming, filtering and hovering over the dashboard visuals.
- Streamlit - Framework for creating web applications that can deploy the interactive analytics dashboard.

5.5 Validation Approach

Analytical findings were validated through multiple methods of cross referencing the results for internal consistency. The main insights were validated/verified against known business domain knowledge and reasonable/common-sense expectations to identify potential artifacts and errors with the data and/or any processing, prior to reporting the final results.

6. Sales Trend Analysis

6.1 Overall Revenue Trajectory

The daily revenue time-series for Afficionado Coffee Roasters indicates that this business has identifiable cyclical revenues that are superimposed on a relatively stable underlying revenue base. The day-to-day fluctuations (daily revenue) that the business experiences are caused by all known retail demand influences (e.g., climate/weather, public holidays, promotional activities). Weekly periodicity is apparent and establishes that there are day-of-week affects as a main source of revenue variability.

By aggregating revenue on a week-by-week basis, we smooth out any short term fluctuations and reveal the longer-term trend in the performance of the business. Some weeks will have higher aggregated revenue totals as a result of higher demand being concentrated around specific promotional periods or certain dates and this allows us to create a recurring revenue baseline from which to quantify any deviation (both positive and negative) in future periods.

6.2 Weekly Aggregation Patterns

As demonstrated through a week-to-week revenue aggregation, the business does not appear to be operating in consistent steady state. Specifically, there are identifiable high performing weeks and low performing weeks. Based on preliminary analysis, the lower performing weeks appear to correspond to holidays, school schedules and local events which reduce the amount of commuter traffic (the primary demand driver for coffee retail) during the morning hours.

According to our week-over-week variance analysis, the highest performing quartile of weeks generates about 35% to 40% more revenue than the lowest performing quartile. This indicates that there is a very significant degree of temporal revenue risk that could be mitigated through better planning.

6.3 Relation of Revenue and Transaction Volume

The correlation between total revenue and the number of transactions is reviewed over the entire observation period. Although these numbers are highly correlated overall, significant disparities exist during specific time frames that are indicative of changes in average basket size, and not simply changes in foot traffic or the number of transactions. When total revenue grows more rapidly than the number of transactions, it represents an opportunity for successful upselling, premiumization of products, and/or increasing purchases made by multiple people at once; all of these are strategic indicators to assist in finalizing the product and pricing strategy..

7. Review of Performance by Day of Week

7.1 Revenue by Day of the Week

The revenue by day of the week analysis shows there is a significant and consistent disparity in both total revenue and the number of transactions over the seven days; the general result from our data is as follows:

Day	Relative Revenue	Transaction Volume	Behavioral Profile
Monday	Moderate	Moderate	Post-weekend reset; commuter-driven purchases
Tuesday	Moderate to High	High	Stable weekday routine purchases
Wednesday	High	High	Mid-week peak; strong habitual demand
Thursday	Moderate to High	High	Pre-weekend buildup; sustained activity
Friday	High	High	Peak weekday; social and commuter traffic
Saturday	Variable to High	High	Leisure visits; longer dwell time
Sunday	Variable to Moderate	Moderate	Family and brunch patterns; late start

7.2 Weekday vs. Weekend Behavioral Differences

The difference in consumer behavior based on whether it is a weekday or weekend presents clear implications for analysts analyzing by days. Weekday usage of stores has a strong correlation to commuter and job-based patterns of demand, providing a very strong and widely occurring Monday–Friday peak in the morning hours from 7 to 9:30 AM, and a smaller secondary peak during the day. In contrast, weekend demand is seen for a much longer/more significant window of time during the morning/afternoon hours, from the leisure & social perspective, and not related to transit-based buying patterns.

Average transaction values for weekends are slightly higher than for weekdays, consistent with what has been documented that leisure shoppers typically purchase more complex items (i.e., higher profit beverages) as well as additional food items. Hence, weekend promotions that include bundled items or higher-end products will likely have more favorable revenue impacts than weekday promotions.

7.3 Average Basket Size Analysis

Average revenue per basket (basket size) has significant fluctuations across the week and is not accounted for transaction volume. As a majority of store sites will exhibit the largest average basket sizes on Saturday and Sunday, it supports the leisure-visitor behavioral hypothesis, and that this provides guidance for stocking periodic inventory at a store location, specifically that greater depth of premium, high-margin products should be put away for weekends; while a store should operate on a weekday to maximize the number of quick-serve (i.e., routine) purchases.

8. Time-of-Day Demand Analysis

8.1 Hourly Demand Distribution

- Transaction data on an hourly basis shows one of the strongest structural findings throughout this entire research endeavor. Afficionado coffee retailers exhibit a considerable concentration of demand in the early morning. Specifically, the bulk of revenue generated each day occurs in a narrow window between about three to four hours in the morning. The hourly distribution of sales at our stores follows a similar pattern across all locations and days, as outlined in the following:
- Early Morning (6:00 a.m.-7:00 a.m.): The first wave of demand for coffee begins to build as the first group of early commuters and pre-work coffee customers arrives.
- Morning Peak (7:00 a.m.-9:30 a.m.): The period of the greatest transaction density and revenue concentration for all weekdays is recorded.
- Mid-Morning (9:30 a.m.-11:00 a.m.): There is a steady decline in demand following the morning peak; however, demand is still well above the average daily demand.
- Midday (11:00 a.m.-1:00 p.m.): Sales performance varies within and across stores during this period. In many (but not all) office-adjacent locations, a substantial number of transactions occur around lunchtime.
- Afternoon (1:00 p.m.-4:00 p.m.): There are very few transactions occurring. The afternoon consists of a low level of sales volume, representing the low season for coffee.
- Evening (4:00 p.m.-6:30 p.m.): In a few specific stores, notably those located in areas with a high student population or transportation hubs, there may be a secondary peak in sales during the start of the last hour of operations.
- Late Evening (After 7:00 p.m.): There will be a decisive reduction in demand for coffee and consequently, there will be minimal coffee trading activity occurring during this period.

8.2 Time-Bucket Performance Summary

Time Bucket	Hours	Demand Level	Strategic Implication
Morning	06:00-11:59	Very High (Primary Peak)	Maximum staff deployment; full inventory readiness

Afternoon	12:00-16:59	Low to Moderate	Staffing reduction opportunity; promotion window
Evening	17:00-21:59	Moderate (Location-Dependent)	Variable staffing based on individual branch profile
Late Hours	22:00-05:59	Minimal	Evaluate operational cost-effectiveness by location

8.3 Heatmap Analysis — Hour x Day Interaction

developing a better understanding of customer behavior by store. Analysis of store-level sales information will help define characteristics of higher performing stores so that lower performing stores can be brought up to or closer to that level. This will benefit all stores within the chain.

At the end of the 12-month reporting period, there was significant difference in performance from top performing stores to bottom performing stores. The differences in total revenue and transaction volume on a store-level basis indicate that customer behavior is not solely due to the individual store's location, but also due to factors such as the chain brand quality and customer service level at that location. Store locations with higher total revenue and transaction volumes have higher levels of foot traffic, are located in areas with higher levels of public transportation access, have greater levels of customer activity (visiting with employees and friends), and have higher levels of customer service than lower performing stores do. Customers' perceptions of the chain brand quality influences customer loyalty, or propensity to return and/or recommend a restaurant to others.

9. Cross-Location Temporal Comparison

9.1 Store-Level Revenue Performance

Through this detailed analysis of customer data by store and the corresponding differences between top and bottom performing stores across all metrics of customer behavior, we can improve our understanding of customer behavior within our chain of stores and develop strategies for improving individual store performance. By considering the various factors that contribute to the performance differences among stores, restaurant owners can implement strategies to increase sales and/or reduce costs in their stores. There are two primary categories that can be used to increase sales: advertising and promotion and/or operational changes. Both of these strategies can create opportunities for the various locations of the chain to serve different markets and customers. All stores are not created equal and there will be differences in customer behavior based on the store's physical location, demographics, labor, competition, etc. Understanding these factors and how they impact overall chain and store performance will ultimately provide each store with an opportunity to maximize its business and become and sustain a successful business.

9.2 Location-Specific Peak Hours

While the morning demand peak is a universal pattern across all Afficionado locations, the precise timing, duration, and magnitude of the peak varies significantly by branch. Key observations include:

- Urban transit-adjacent stores exhibit the sharpest and earliest peaks (7:00-8:30 AM), with rapid demand dropoff after 9:30 AM.
- Stores in business park or office campus environments show a broader morning peak extending to 10:00 AM, reflecting flexible arrival times among professional workers.
- Locations near educational institutions show a distinct dual-peak pattern, with the morning peak supplemented by a significant late-afternoon cluster (3:30-5:30 PM) corresponding to post-class demand.
- Residential neighborhood stores demonstrate the most evenly distributed demand profile, with the morning peak less pronounced and afternoon demand relatively higher than other location types.

9.3 Implications for Location-Differentiated Strategy

A unique approach to each store's demand is needed to implement a successful uniform chain-wide plan, requiring the development of unique methods for staffing, ordering inventory, and timing promotions at the store based on their specific demand profile. For example, a campus store and a transit hub store have fundamentally different ways of operating even if their actual monthly total sales are the same.

10. Dashboard Development

10.1 Design Philosophy

The dashboard that allows users to interactively explore and obtain analytical results is built upon the idea of data "analytical accessibility." That is, results obtained from extensive analysis of data will be accessible and usable by non-data scientists. The dashboard serves as an exploratory analytical environment for experienced analysts, and as a tool to support the decision-making of store managers and chain executives with limited experience in using technology.

10.2 Technology Stack

- Streamlit - application framework for rendering an interactive user interface and specifying filtering logic
- Plotly - interactive charting library that provides the capability for users to hover, zoom, pan and filter charts interactively
- Pandas - receives current user filters and aggregates the data in real-time
- NumPy - optimizes performance to quickly perform calculations on large datasets

10.3 Core Dashboard Modules

Module 1: Overall Sales Trend Dashboard

Displays daily and weekly revenue trend lines for the entire observation period and supports selection of date range and toggling of level of aggregation (daily, weekly, monthly). Includes a trailing 7-day moving average to reduce short-term fluctuations in the data.

Module 2: Day-of-Week Performance Charts

This chart shows revenue, transaction counts, and average baskets on a weekly basis (Monday–Sunday) per location and date range. Weekday/weekend breakdowns can be toggled on and off.

Module 3: Hourly Demand Heatmaps

Heatmaps showing demand intensity over days/hours and daytypes would help to visualize how demand changes during the various hours of the day, week and/or month. The color intensity will indicate either the transaction volume or revenue. The interactive feature of being able to hover over a cell to display its exact volume or revenue would also be beneficial. An alternative view would be to aggregate the time frames into buckets.

Module 4: Cross-Location Comparison Panels

This module provides side-by-side comparisons of the same performance measures for each store, including total store revenues; identifying peak hours for each location; and comparing weekday to weekend revenues. It allows for selection of specific stores to focus on comparing.

10.4 User Interaction Capabilities

- Store location filter – a single or multi-select dropdown option allowing users to filter all views on the dashboard by specific branches they selected.
- Day of week selection – checkboxes to include or exclude any day(s) of the week in all calculations.
- Hour Range Slider – a continuous slider widget used to restrict data analysis to a particular time period.
- Revenue or Quantity (toggle) – a method of switching between revenue based metrics or transaction based metrics in any chart on the dashboard.
- Date Range Picker – a calendar-based site where users can select dates to isolate the analysis to a specific business period.
- Export Functionality – allows a user to download both filtered data tables and chart images for reporting purposes.

11. Key Findings

The following represent the primary findings of the Sales Trend and Time-Based Performance Analysis for Afficionado Coffee Roasters, synthesized from all analytical modules described in this paper:

#	Key Finding	Evidence Basis
1	Morning hours (7:00-9:30 AM) represent the highest demand period, accounting for a disproportionate share of daily revenue across all store locations and days of the week.	Hourly transaction distribution; time-bucket aggregation
2	Sales are highly time-dependent and follow predictable, cyclically repeating patterns that are stable across the observation period.	Week-over-week trend analysis; day-of-week aggregation
3	Weekend customer behavior is fundamentally different from weekday behavior in timing, average basket size, and product preference distribution.	Weekday/weekend split analysis; hourly demand curves
4	Store performance varies significantly by location, with location type explaining meaningful portions of demand pattern variance.	Cross-location peak hour analysis; revenue ranking by branch
5	Revenue and transaction volume are strongly correlated overall but diverge during promotional periods and on weekends, signaling basket-size effects.	Revenue vs. transaction count dual-metric analysis
6	The afternoon period (12:00-4:00 PM) represents a consistent, universal low-demand window across all locations, representing an optimization opportunity.	Heatmap analysis; time-bucket revenue share decomposition

7	An evening secondary demand peak exists at education-adjacent and transit-hub locations, but is absent at residential neighborhood stores.	Location-specific hourly demand profiles

12. Recommendations

The following recommendations are derived directly from the analytical findings and are organized by functional domain for ease of implementation planning:

12.1 Workforce Management

- Early morning staffing concentration - Increase staff by 25 to 40% during the peak period of Monday through Friday between 7:00 AM - 9:30 AM at all locations served by Transit or Offices. This increase will help ensure that service speed and capacity are aligned with peak demand periods.
- Reduce late afternoon labor - Decrease floor labor during non-peak hours of 12:00 noon - 4:00 PM by 20 to 30%. Reallocate these hours to pre-peak setup and cleaning. Service levels should not fall below acceptable standards.
- Develop This is a separate weekend specific staffing model that aligns all resources into the peak windows of 8:30 AM - 11:30 AM. This weekend staffing model will have a flatter profile as opposed to the steeper profile that exists on weekdays.
- Location specific scheduling - Instead of applying a single chain-wide template, develop separate shift models for each of the four archetypes: Transit, Campus, Office, and Residential.

12.2 Inventory Planning

- Fresh product daily staging - Establish timeframes for receipt of fresh ingredients and complete setup of the store by 6:30 AM during weekdays and 7:30 AM during weekends to ensure that all products are available for purchase when the store opens to customers at peak demand.
- Adjust ordering cycle to demand - Align weekly ingredient orders to the weekly demand profile and order more product for days with high demand and less for those with lower statistical levels of demand.
- Build depth of stock for premium beverages and/or food to take advantage of the proven increased baskets on Saturday and Sunday.

- Afternoon markdown strategy — consider targeted discounts on perishable pastry items during the 2:00-4:00 PM low-demand window to reduce waste while stimulating afternoon traffic.

12.3 Marketing and Promotional Strategy

- Demand stimulation campaigns in the afternoon — create marketing campaigns specifically designed to stimulate demand during the 12:00 PM - 4:00 PM timeslot through the use of loyalty multiplier rewards, food-and-coffee bundles, and limited time promotional pricing.
- Weekday loyalty program optimization — capitalize on the strong habitual purchase behavior that exists during weekday mornings and enhance participation in loyalty programs by utilizing this time frame as a way of reinforcing loyalty to the brand.
- Weekend experiential marketing — utilize the weekend's leisure-based customers to develop and implement promotional experiences in-store (sample flights, barista demonstration classes, community events) for Saturday and Sunday mornings only.
- Location-based promotional calendars — develop a uniquely tailored promotion plan for each location type based on its individual demand patterns rather than roll out a uniform corporate-wide promotion plan.

12.4 Technology and Analytics Infrastructure

- Implementation of interactive dashboard tools — provide store managers and regional operations teams with Streamlit dashboards to use as a regular management tool in weekly performance reviews.
- Implementation of KPI tracking cadence — establish weekly automated reporting which provides; peak-hour sales volume by day and location; and revenue received by location, relative to other locations.
- Real-time monitoring of operations — connect the Streamlit dashboard to POS transaction feeds so; daily staffing adjustments can be made during the day based on the volume of customers at each location.
- An expanded analytical forecasting model — include external information (e.g., weather, local special events, holidays) in your forecasting model to develop an accurate demand forecast..

13. Conclusion

This paper presents an extensive methodological analysis of the sales trends and time-based performance patterns of Afficionado Coffee Roasters. Through the systematic application of data cleaning, the engineering of time-based features, multi-dimensional statistical analysis, and interactive visualizations, we have been able to transform raw transactional data into a rich and useful set of actionable business intelligence.

The key finding of this study is that coffee retail sales are extremely time concentrated, with the majority of daily revenue occurring during the morning hours, and that there are significant variations in the time patterns by day of the week and by location. As such, there are important and immediate implications for staffing, inventory, and marketing strategies at all Afficionado locations.

The identification of a consistent low demand window in the afternoon provides an opportunity for operational improvement: the same staffing resources that are utilized during this period could be applied to generating greater value if they were reallocated to support peak demand or to prepare for service. Likewise, the documented differences in weekday demand, which are driven by commuting, versus weekend demand, which is driven by leisure, suggest that the operational model for Afficionado should be different from the current uniform operational model.

No one model of operation can operate in a way that can serve all of the Afficionados' branches at the same time. Each of the four types of location demand archetypes (transit hubs, office parks, campus-adjacent, residential) has a time of day and week where it is in high demand, and therefore these demand archetypes require unique schedules, inventory levels, and promotional strategies.

An interactive Streamlit dashboard was designed as a part of the work done in this study to provide an open and continually available, stakeholder-ready decision support tool. The dashboard will allow users to explore and view real-time data they've been trained to use without any technical background. The dashboard has helped democratize organization access to data while supporting the organization's cultural shift to an evidence-based operational management approach.

The analytical framework developed in the study can be developed in three different ways in the future: 1. by developing predictive demand forecasting models that utilize data from external demand drivers, 2. by building the capability to receive live POS systems data to monitor operational metrics in real time, and 3. by broadening product-level analysis to identify high-margin products that exist within specific time periods and utilize the product margins to develop targeted menu engineering strategies.

It is clear that applying systematic, time-based analytics to the existing rich transactional data held by Afficionado Coffee Roasters will provide significant and measurable improvements in terms of efficiencies within their operations, revenue generation, and the quality of their customers' experience. If the recommendations provided in this report are executed systematically and based on location of implementation, they will provide a high return on investment for developing data-driven retail management capabilities.

— End of Research Paper —

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